Chemical Kinetics

Introduction

Kinetics involves the study of progress of a process from one state to another during a physical and/or chemical change. It starts with the measurement of rates of reactions, followed by the determination of the rate laws at the macroscopic level. Finally, the rate law is used to understand the mechanism of reaction at the microscopic level.

Thermodynamics is concerned with only the initial and final states, whereas kinetics deals with what happens to a particular process at any time between two states. Thermodynamics helps us to find out the amount of product at equilibrium or on completion of a reaction. On the other hand, kinetics helps us to find the yield at any given time.

Some reactions are slow whereas others are fast. Some reactions happen only at high temperature. The knowledge of this can be used to our advantage. The degradation of paint and plastic should be slow, whereas an air–fuel mixture should burn rapidly at ignition. Burning of cooking gas should be fast whereas rusting of iron needs to be slowed down. Rate of oxidation of a fuel should be negligible at room temperature, but it should be very fast at the high temperature at which an engine operates.

Kinetic studies can be used to know the following.

- Amount of product or reactant: The rate law and its integrated form can be used to know the amount of product or reactant at any time. It also helps in finding the time required to obtain the maximum amount of product.
- Stability and rate of drug metabolism: A drug's stability is measured over time at typical room temperature and humidity conditions. This helps in deciding its expiry date. Moreover, the knowledge of its metabolic rate is essential in prescribing the interval of drug intake.
- Degradation/decomposition rate: Knowledge of the rate of degradation is important not only in drug industry, but also in bioremediation, degradation of environmental pollutants, deterioration of food, etc.
- Rate of product formation at different temperatures: Temperature can have dramatic effects on the rate of a reaction. Kinetics can help us quantify the effect of temperature on the reaction rate.

- Mechanism of reaction, microscopic nature of a reaction: One of the most important uses of kinetics is to know the mechanism of reaction. It helps us to devise alternative strategies for obtaining products in higher yield and/or shorter time.
- Effect of solvent on the reaction rate: Understanding the mechanism of a reaction helps us to know how the polarity and nature of solvent (protic or aprotic) can affect the rate of reaction.
- Slow and fast reactions: Some reactions are instantaneous, and some are slow. Kinetics enables us to know the rate of reaction.
- Structure and nature of transition state: Knowledge of potential energy surface of reactants can help us to know the structure and nature of transition state.
- Role of catalysis: One of the most important applications of kinetics is in the field of enzyme catalysis. Kinetics helps us to understand the mechanism of their action.
- Radiometric dating: The age of ancient relics, such as paintings, dinosaur fossils, and remains from ancient civilizations can be obtained by kinetic studies involving radiometric dating.

In order to understand and predict the behaviour of a physical or chemical process, knowledge of both thermodynamics and kinetics is required.

Important Concepts in Kinetics

Stoichiometric coefficient of a reactant/product: Stoichiometry represents the quantitative relationship between the number of moles of reactants and products based on the law of conservation of mass. It is negative for reactant(s) and positive for product(s).

For the reaction:

$\mathrm{N_2(g) + 3H_2\ (g) \leftrightharpoons 2NH_3(g)}$,

the balanced equation suggests that one mole of $\mathrm{N_2}(g)$ reacts with three moles of $\mathrm{H_2}\ (g)$ to give two moles of $NH_3(g)$ if all of the nitrogen is used up. Thus, in this reaction, the stoichiometric coefficients of reactants/products $N_2(g)$, $H_2\ (g)$ and $NH_3(g)$ are -1, -3, and 2, respectively.

Kinetic reaction profile: This is obtained by measuring the concentration of one of the reactant or product as a function of time during the course of a reaction.

Rate of reaction: Every chemical or physical process takes place at its own speed. The speed of a chemical reaction is quantitatively measured in terms of the rate of the reaction. Rate of the reaction (ϑ) is defined as the change in the extent of reaction (ξ) with time (t) per unit volume (V).

$$\vartheta = \frac{1}{V}\frac{d\xi}{dt} \qquad (6.1)$$

The infinitesimal change in number of moles of 'i' th reactant or product (dn_i) is given by

$$dn_i = \nu_i d\xi$$

where ν_i is the stoichiometric coefficient of 'i' th species (reactants or products). It is negative for reactants and positive for products.

$$\vartheta = \frac{1}{V}\frac{1}{\nu_i}\frac{dn_i}{dt} = \frac{1}{\nu_i}\frac{dC_i}{dt} \qquad (6.2)$$

where C_i (= n_i/V)is the concentration of 'i' th reactant or product and volume is assumed to be constant.

For a reaction

A+2B→3C+D

$$\vartheta = \frac{d[D]}{dt} = \frac{1}{3}\frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{1}{2}\frac{d[B]}{dt}$$

$$\left.\frac{d[D]}{dt}\middle/\frac{d[C]}{dt}\right. = 1/3$$

Thus, the rate at which any reactant/product is consumed or formed is proportional to its stoichiometric coefficient. The rate of a reaction is zero at equilibrium. It is important to optimize the rate of a reaction to get maximum yield in less time.

Determination of rate of a reaction: The rate of a reaction can be obtained by looking at the rate of change of concentration of one of the reactant or product with time *t*. The slope of the plot of concentration versus time at any given instant will give the rate of change of concentration at that time.

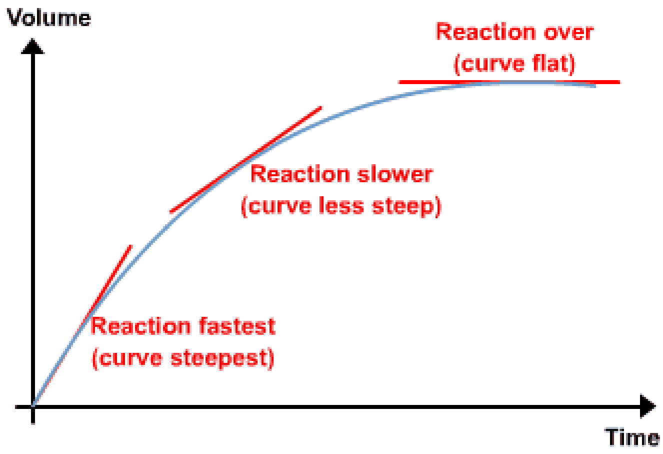

Fig. 6.1: Concentration of product as a function of time can be used to calculate the rate of reaction.

Factors affecting the rate of a reaction:

Concentration of reactants: Concentration of reactants, in general, increases the rate of reaction. However, it may also not affect the rate of a reaction for example, concentration of iodine does not affect the rate of reaction of iodination of acetone in the presence of H^+. The dependence of reaction rate on concentration of the reactant is given by the rate law. A type of reactant, called inhibitor, decreases the rate of a reaction.

Nature of reactants: Nature of reactants greatly influences the rate of reaction; for example, the rate of reaction of a nucleophilic substitution at aliphatic halide (R−X) will differ for different R and X. The rate of a nucleophilic substitution reaction follows the order: R−Cl < R−Br < R−I since the bond gets weaker from R−Cl to R−I.

Stability of intermediates: The stability of intermediates may also affect the rate of reaction. In SN^1 nucleophilic reaction, the rate of reaction increases with increase in the stability of intermediate.

Nature of solvent: The solvent can stabilize the intermediates and can affect the rate of a reaction. A polar solvent stabilizes the carbocation formed and hence increases the rate of a reaction where carbocation is involved (in S_N1 substitution and E1 elimination).

Temperature: In most cases, rate of a reaction increases with increase in the temperature. The increase in the reaction rate can be explained by Arrhenius hypothesis (explained later). In most cases, the rate gets doubled with every 10 K increase in temperature.

Surface area of reactant: Increase in surface area of reactant leads to increase in the rate of a reaction. Thus, the rate of reaction is much faster with a finely divided reactant than with its larger pieces.

Catalysts: They increase the rate of a reaction without themselves undergoing any change. .They do it by decreasing the activation energy (explained later).

Photochemical radiations: The rate of some reactions, called photochemical reactions, increases when the reactants are irradiated by electromagnetic radiation of certain wavelength.

Law of mass action (rate law): The rate of a reaction (ϑ) depends on the concentration of the reactants (A, B…..). This is known as the law of mass action.

$\vartheta = f([A], [B], \dots \dots ..)$

$$\vartheta = k[A]^a[B]^b \qquad (6.3)$$

where k is the rate constant and is independent of concentration. $[A]$ and $[B]$ are the concentration of reactants A and B, respectively; and a and b are numbers obtained from the experimental methods. It should not be determined by simply looking at the stoichiometry of the balanced equation.

Order of a reaction: It is obtained experimentally by measuring the dependence of the rate of a reaction on the concentration of reactants. The overall order of the reaction is equal to the sum of exponents of the concentration terms in the rate law.

$\vartheta = k[A]^a[B]^b$

Order of the reaction = a+b

Order of the reaction with respect to A = a

Order of the reaction with respect to B = b

The unit of k will depend on the order of reaction.

Elementary and composite reactions: Some of the reactions take place in one step whereas others proceed via multiple steps. A reaction taking place in one step is called an elementary reaction whereas a reaction which involves more than one elementary step is known as composite reaction.

Molecularity of a Reaction: Molecularity is defined only for an elementary reaction. It is the number of reactant molecules that takes part in an elementary reaction. Thus, for an elementary reaction, A$\rightarrow P$, the molecularity is one. For an elementary reaction, $A + B \rightarrow P$, molecularity is 2. Molecularity is same as the overall order of a reaction for an elementary reaction.

For a composite reaction, the molecularity is the total number of reactants taking part in the rate determining step of a reaction. Molecularity is always an integer, whereas order can be fraction or zero.

Determination of Rate law:

Reactions involving single reactant

The following methods can be used to find the order of reactions involving a single reactant.

Differential Method: It can be used to determine the order with respect to a particular reactant. If the reaction is of n^{th} order with respect to a reactant A, then

$$\vartheta = k[A]^n \qquad (6.4)$$

$$\frac{\vartheta}{[A]^n} = k \qquad (6.5)$$

The order of the reaction with respect to A will be the value of n at which $\frac{\vartheta}{[A]^n}$ is constant.

A more simplified approach can be obtained as follows:

Taking log on both sides, we get

$$ln\vartheta = ln(k[A]^n)$$

$$ln\vartheta = lnk + nln[A] \qquad (6.6)$$

Thus, the plot of $ln\vartheta$ versus $ln[A]$ will be a straight line and the slope of the plot will be equal to the order of reaction.

Integrated Rate Laws: Rate laws are differential equation. They can be integrated to obtain the integrated rate law. These laws can be used to determine the order of a reaction**.** They can be derived for different order as follows:

Integrated rate law for a zero order reaction:

A reaction, $A \rightarrow P$, is of zero order if rate of reaction is independent of concentration of reactant A as shown below :

$$-\frac{d[A]}{dt} = k$$

The unit of k for a zero order reaction is same as the rate of the reaction, that is, mol $L^{-1}s^{-1}$

$$-d[A] = kdt$$

Integrating between time t=0 to t=t, we get integrated rate law for zero order

$$-\int_{[A]_0}^{[A]} d[A] = \int_{t=0}^{t=t} kdt$$

$$[A]_0 - [A] = kt$$

$[A]_0$ is the concentration of A at t=0, $[A]$ is the concentration of A at time t.

Examples of zero order reaction: A zero order reaction is generally observed in reactions involving a surface or a catalyst involved in the reaction is saturated with reactant.

(i) Decomposition of N_2O on hot platinum surface: In this reaction, platinum surface is saturated with N_2O and the rate of reaction does not depend on the concentration of N_2O.

(ii) Iodination of acetone in the presence of H^+ : The rate of reaction is zero order with respect to iodine.

Integrated rate law for First order reaction: A reaction is of first order if the rate of the reaction is proportional to the concentration of the reactant.

$$-\frac{d[A]}{dt} = k[A]$$

$$-\frac{d[A]}{A} = kdt$$

Integrating between time t=0 to t=t, we get integrated rate law for first order.

$$-\int_{[A]_0}^{[A]} \frac{d[A]}{A} = \int_{t=0}^{t=t} kdt$$

$$ln\frac{[A]_0}{[A]} = kt$$

$$ln[A] = ln[A]_0 - kt$$

For a first-order reaction, a plot of ln $[A]_t$ vs t will yield a straight line with a slope of $-k$. Unit of k is s^{-1}.

Examples of first order reactions

(i) Decomposition of H_2O_2: The rate of reaction depends on concentration of H_2O_2.

(ii) Inversion of cane sugar: The rate of reaction depends on concentration of sugar.

Integrated rate law for second order reaction: A reaction is of second order if the rate of the reaction is proportional to the square of the concentration of reactant.

$$-\frac{d[A]}{dt} = k[A]^2$$

$$-\frac{d[A]}{[A]^2} = kdt$$

Integrating between time t=0 to t=t

$$-\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^2} = \int_{t=0}^{t=t} kdt$$

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

So if a process is second-order in A, a plot of 1/[A] vs *t* will yield a straight line with a slope of *k*.

Examples of second order reactions

(i) Decomposition of NO_2 to NO

(ii) Decomposition of N_2O to N_2

(iii) Saponification of ester

For *n*th order reaction

$$-\frac{d[A]}{dt} = k[A]^n$$

$$-\frac{d[A]}{[A]^n} = kdt$$

Integrating between time t=0 to t=t

$$-\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^n} = \int_{t=0}^{t=t} kdt$$

$$\frac{1}{[A]^{n-1}} - \frac{1}{{[A]_0}^{n-1}} = kt$$

Table 1: Integrated rate equation and linearized plot for different orders of reaction

Order of reaction	Integrated rate equation	Linear plot
Zero order	$[A]_0 - [A] = kt$	$[A]$ vs t
First order	$ln\frac{[A]_0}{[A]} = kt$	$-ln[A]$ vs t
Second order	$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$	$\frac{1}{[A]}$ vs t
nth order	$\frac{1}{[A]^{n-1}} - \frac{1}{{[A]_0}^{n-1}} = (n-1)kt$	$\frac{1}{[A]^{n-1}}$ vs t

Half-Life method: Half-life is defined as the time required for one-half of a reactant to react. Because [A] at $t_{1/2}$ is one-half of the original [A], $[A]_t = 0.5\ [A]_0$. The dependence of half-life on reactant concentration is different for reactions of different order. Thus, this fact can be exploited to find the order of the reaction.

Half-life of a first order reaction:

$$ln\frac{[A]_0}{0.5[A]_0} = kt_{\frac{1}{2}}$$

$$ln2 = kt_{\frac{1}{2}}$$

$$t_{\frac{1}{2}} = \frac{0.693}{k}$$

Thus, half-life of the reactant does not depend on its concentration in a first order reaction.

Half-life of a second order reaction:

$$\frac{\mathbf{1}}{\mathbf{0.5}[A]_0} - \frac{\mathbf{1}}{[A]_0} = kt_{\frac{1}{2}}$$

$$\frac{\mathbf{2}}{[A]_0} - \frac{\mathbf{1}}{[A]_0} = kt_{\frac{1}{2}}$$

$$\frac{\mathbf{1}}{[A]_0} = kt_{\frac{1}{2}}$$

$$t_{\frac{1}{2}} = \frac{\boldsymbol{1}}{\boldsymbol{k}[A]_0}$$

Thus, half-life of reactant is inversely proportional to its initial concentration in a second order reaction.

Half-life of an nth order reaction:

$$\frac{1}{[A]^{n-1}} - \frac{1}{[A]_0{}^{n-1}} = (n-1)kt_{\frac{1}{2}}$$

$$\frac{1}{(0.5[A]_0)^{n-1}} - \frac{1}{[A]_0{}^{n-1}} = (n-1)kt_{\frac{1}{2}}$$

$$\frac{2^{n-1}}{([A]_0)^{n-1}} - \frac{1}{[A]_0{}^{n-1}} = (n-1)kt_{\frac{1}{2}}$$

$$\frac{2^{n-1}-1}{([A]_0)^{n-1}} = (n-1)kt_{\frac{1}{2}}$$

$$t_{\frac{1}{2}} = \frac{2^{n-1}-1}{(n-1)k([A]_0)^{n-1}}$$

Thus, half-life of the reactant is inversely proportional to $([A]_0)^{n-1}$ in an nth order reaction.

Table 6.2 summarizes the inference of order from the concentration dependence of half-life.

Table 6.2: Inference of order from the concentration dependence of half-life

Concentration dependence of half-life	Order of reaction
$t_{\frac{1}{2}} \propto [A]_0$	Zero order
$t_{\frac{1}{2}} \propto ([A]_0)^0$	First order
$t_{\frac{1}{2}} \propto ([A]_0)^{-1}$	Second order
$t_{\frac{1}{2}} \propto ([A]_0)^{n-1}$	nth order

Reactions involving more than one reactants:

Isolation method: It is a very useful method to determine the order of reaction with respect to a particular reactant when more than one reactant is involved. In this, the other reactants are taken in large excess such that their concentration remains virtually the same during the entire course of the reaction. Thus, in this case, the rate of the reaction will essentially depend only on one reactant and thus the methods used for determining the rate order for one reactant reactions can be applied.

Suppose a reaction involves two reactants, A and B.

$$\vartheta = k[A]^{\alpha}[B]^{\beta}$$

To determine the order of reaction with respect to A, we will take B in large excess. Thus, concentration of B will remain virtually the same throughout the reaction.

$$[B] \approx [B]_0$$

where $[B]_0$ and $[B]$ are initial concentration of B and concentration of B at a given time, respectively. Thus

$$\vartheta = k[A]^{\alpha}[B]_0^{\beta}$$

$$\vartheta = k\prime[A]^{\alpha}$$

Thus, rate of reaction is dependent on only concentration of A and thus order with respect to A can be measured as measured in case of reactions involving single reactant. Similarly, order of reaction with respect to reactant B can be measured by taking A in large excess.

Method of initial rate: This method is based on the measurement of rate at the initial stages of the reaction for different concentrations of only one reactant while keeping the concentration of others constant.

Suppose a reaction involves two reactants A and B.

$$\vartheta = k[A]^{\alpha}[B]^{\beta}$$

The rate of reactions is determined at initial stage of the reaction where $[A]$~ $[A]_0$ and $[B]$~$[B]_0$.

$$\vartheta = k{[A]_0}^{\alpha}{[B]_0}^{\beta}$$

Now, to get the order of reaction with respect to A, change the initial concentration of A keeping the initial concentration of B constant. Suppose that initial concentration of A is changed to $[A]_0'$

$$\vartheta' = k{[A]_0'}^{\alpha}{[B]_0}^{\beta}$$

$$\frac{\vartheta'}{\vartheta} = \left(\frac{[A]_0'}{[A]_0}\right)^{\alpha}$$

If the reaction is of first order with respect to A

$$\frac{\vartheta'}{\vartheta} = \frac{[A]_0'}{[A]_0}$$

If the reaction is of second order with respect to A

$$\frac{\vartheta'}{\vartheta} = \left(\frac{[A]'_0}{[A]_0}\right)^2$$

Similarly, the order of reaction with respect to B can be obtained by changing $[B]_0$ and keeping $[A]_0$ constant.

In this method, there is no need to take the other reactants in large excess. However, this method may not work for fast reactions.

Integrated rate law in terms of physical parameter: Monitoring of concentration with time is difficult and in some cases not possible. However, monitoring of a physical property that is proportional to concentration, can help us in obtaining the order of reaction and thereby the rate constant. For a gaseous reaction in a closed container, the pressure of gaseous reactants and products at a given temperature T is proportional to their number of moles. Thus, measurement of the total pressure during the course of reaction can help in obtaining integrated rate law in terms of a physical parameter.

For a reaction, $2\ NO_2\ (g) \rightarrow N_2O_4(g)$

Total pressure will be given by

$P_T = \frac{n_{NO_2}RT}{V} + \frac{N_2O_{4P}RT}{V} = kn_A + kn_P$

$P_T \propto (n_A + n_P)$

and k=RT/V will be same for reactant or product, since both are at same temperature and have same volume.

Let us consider the decomposition of ammonia in a closed container at certain temperature. Ammonia breaks into nitrogen gas and hydrogen gas. Let $P_0, P_t\ and\ P_\infty$ be the initial pressure, pressure at time t, and final pressure of gas in the container, respectively. Pressure is proportional to the number of moles at constant volume and temperature.

$$NH_3(g) \quad \rightarrow \quad \frac{1}{2}N_2(g)+\frac{3}{2}H_2(g)$$

@ time $t=0$	a	0	0	$P_0 = k_1 a$
@ time $t=t$	(a - x)	$x/2$	$3x/2$	$P_t = k_1(a+x)$
@ time $t=\infty$	0	$a/2$	$3a/2$	$P_\infty = k_1 2a$

$$P_t - P_\infty = -k_1(a-x)$$
$$P_0 - P_\infty = -k_1 a$$

$$\frac{P_t - P_\infty}{P_0 - P_\infty} = \frac{(a-x)}{a}$$

Any other physical parameter can also be used to get the ratio of (a−x)/a. Thus, optical rotation can be used to obtain this ratio if the reaction involves chiral reactant(s) or product(s). Ionic conductivity can be used if we are dealing with ionic reactions. However, these parameters are a bit different than pressure. In case of pressure, 1 mole of any of the reactant or product gases exerts the same pressure in a closed container and thus proportionality constant between pressure and number of moles is same for any of the reactants or products. On the other hand, optical rotation and ionic conductivity of reactant/product is proportional to its concentration, the proportionality constant for each reactant and product will be different.

For the reaction,

Sucrose → Glucose + Fructose

Total optical rotation = $k_{sucrose}n_{sucrose} + k_{glucose}n_{glucose} + k_{Fructose}n_{Fructose}$

Here, the value of k will be different for sucrose, glucose, and fructose, since optical rotation per mole of sucrose, glucose and fructose is not the same.

Consider a reaction, $A \rightarrow P$. Let Y_0, Y_t *and* Y_∞ be the values of a certain physical parameter at time zero, at time t , and at the completion of reaction, respectively. The physical parameter of a species is proportional to the number of moles of that species and k_1 and k_2 are the rates of proportionality for reactant and product, respectively.

	A	$\rightarrow$	P	
@ time $t = 0$	a		0	$Y_0 = k_1 a$
@ time $t = t$	(a - x)		x	$Y_t = k_1(a - x) + k_2 x$
@ time $t = \infty$	0		a	$Y_\infty = k_2 a$

$$Y_t - Y_\infty = k_1(a - x) - k_2(a - x) = (k_1 - k_2)(a - x)$$
$$Y_0 - Y_\infty = (k_1 - k_2)a$$

$$\frac{Y_t - Y_\infty}{Y_0 - Y_\infty} = \frac{(a - x)}{a}$$

In fact a/(a–x) is related to physical parameter in similar way, independent of the order of the reaction.

For another reaction,

	A	+	3B	$\rightarrow$	2C	+	D	
@ time $t = 0$	a		b		0		0	$Y_0 = k_1 a + k_2 b$
@ time $t = t$	(a - x)		(b - 3x)		2 x		x	$Y_t = k_1(a - x) + k_2(b - 3x) + k_3(2x) + k_4(x)$
@ time $t = \infty$	0		(b - 3a)		2 a		a	$Y_\infty = k_2(b - 3a) + k_3(2a) + k_4 a$

$$Y_t - Y_\infty = -(a - x)(2k_3 + k_4 - k_1 - 3k_2)$$
$$Y_0 - Y_\infty = -a(2k_3 + k_4 - k_1 - 3k_2)$$

$$\frac{Y_t - Y_\infty}{Y_0 - Y_\infty} = \frac{(a - x)}{a}$$

This relation can be used to obtain integrated rate law for reactions of different order

For a first order reaction

$$ln\frac{[A]_0}{[A]} = ln\frac{a}{(a - x)} = kt$$

$$ln\frac{[A]_0}{[A]} = ln\frac{Y_0 - Y_\infty}{Y_t - Y_\infty} = kt$$

Thus, a plot of $ln\frac{Y_0-Y_\infty}{Y_t-Y_\infty}$ versus t will be a straight line for a first order reaction and the slope will give the value of first order rate constant.

For a second order reaction

$$\frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$\frac{[A]_0}{[A]} - 1 = kt[A]_0$$

$$\frac{[A]_0}{[A]} = kt[A]_0 + 1$$

$$\frac{\boldsymbol{Y_0} - \boldsymbol{Y}_\infty}{\boldsymbol{Y_t} - \boldsymbol{Y}_\infty} = kt[A]_0 + 1$$

Thus, a plot of $\frac{\boldsymbol{Y_0}-\boldsymbol{Y}_\infty}{\boldsymbol{Y_t}-\boldsymbol{Y}_\infty}$ versus t will be a straight line for a first order reaction and the slope/$[A]_0$ will give the value of second order rate constant.